

Live Weather Input

Latitude

41.8781

-

+

Longitude

-87.6298

-

+

Fetch Live Weather

Supply Chain AI - Adaptive Routing Dashboard

Live freight indicators, weather disruption signals, and reinforcement learning policy comparison

Live Economic and Weather Signals

Truck Transportation PPI

196.29

Truckload Index

185.24

Local Freight Index

192.00

Stored Weather Locations

4

User-Selected Live Weather Forecast

Nearest location from coordinates: Chicago, IL

Location

Chicago, IL

Temperature

73 F

Forecast

Showers An...

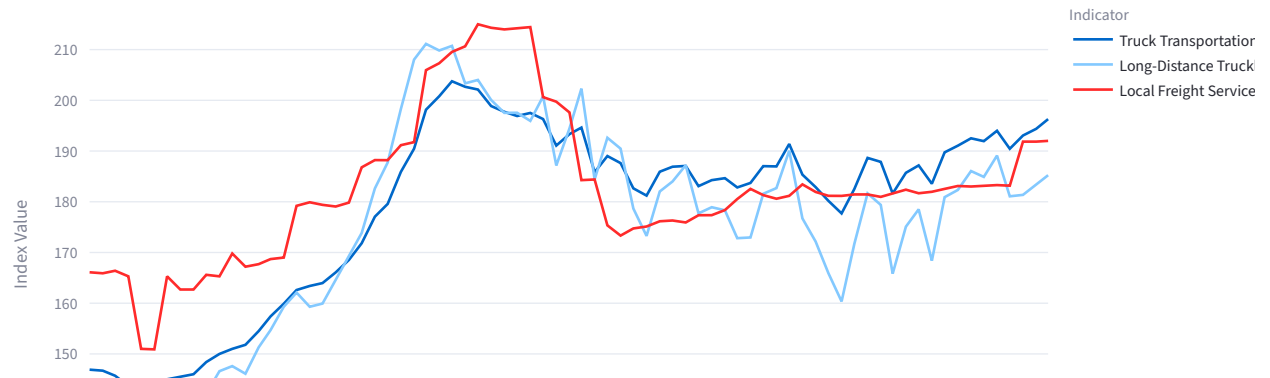
Wind

15 mph SSW

	city	state	latitude	longitude	forecast_name	temperature	temperature_unit	wind_speed	wind_direction	short_forecast
0	Chicago	IL	41.8781	-87.6298	This Afternoon	73	F	15 mph	SSW	Showers And Thunderstorms Lik

Freight and Trucking Economic Indicators

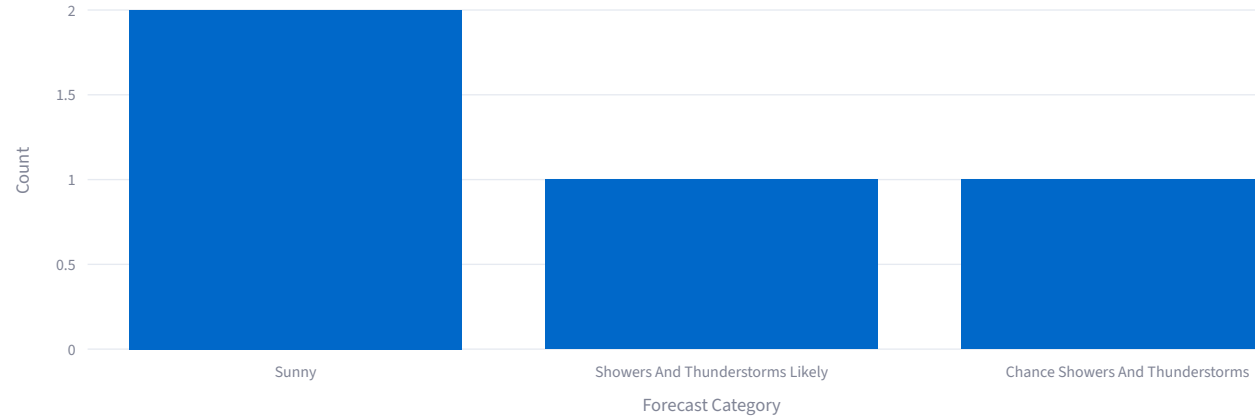
Freight and Trucking Economic Indicators Over Time



Stored Weather Risk Snapshot

	latitude	longitude	forecast_name	temperature	temperature_unit	wind_speed	wind_direction	short_forecast	is_daytime
0	41.8781	-87.6298	This Afternoon	73	F	15 mph	SSW	Showers And Thunderstorms Likely	☒
1	32.7767	-96.797	Today	81	F	10 to 15 mph	S	Chance Showers And Thunderstorms	☒
2	33.4484	-112.074	Today	86	F	0 to 5 mph	SSE	Sunny	☒
3	33.749	-84.388	Today	87	F	5 to 10 mph	W	Sunny	☒

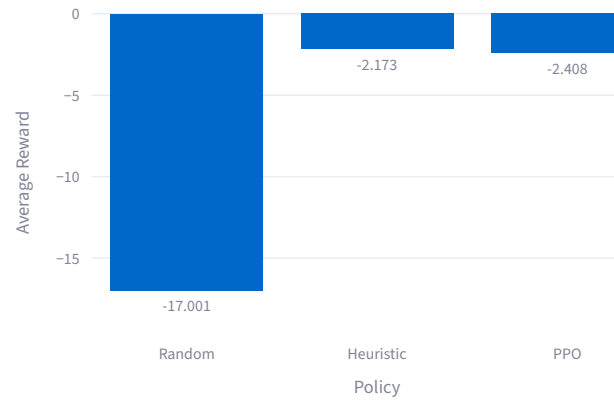
Current Forecast Categories Across Stored Logistics Hubs



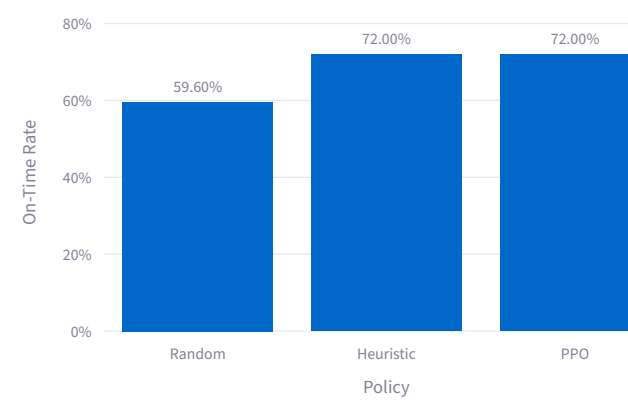
Routing Policy Performance

	model	avg_reward	avg_cost	avg_on_time_rate	avg_disruption_score
0	Random	-17.0008	0.6051	0.596	0.5151
1	Heuristic	-2.1731	0.6307	0.72	0.4068
2	PPO	-2.4078	0.6629	0.72	0.3731

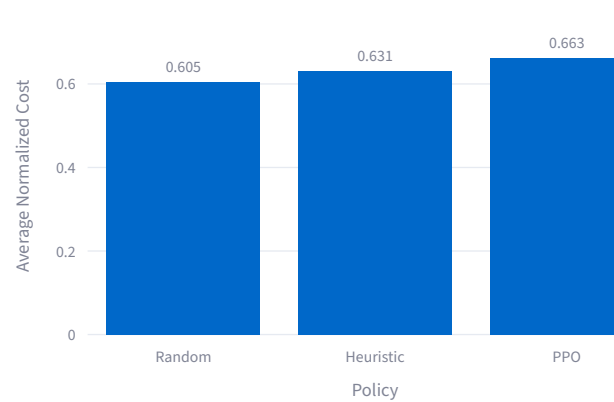
Reward Comparison Across Routing Policies



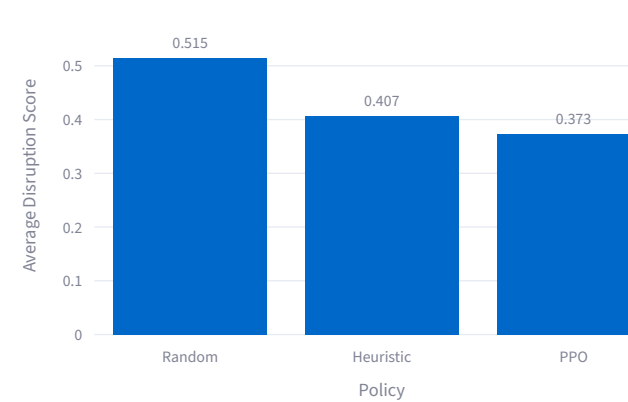
On-Time Delivery Rate Comparison



Operational Cost Comparison



Disruption Exposure by Policy



Key Insight

- **Best overall reward:** Heuristic
- **Lowest operational cost:** Random
- **Highest on-time rate:** Heuristic
- **Lowest disruption exposure:** PPO

This dashboard shows that routing policies behave differently under uncertainty. Heuristic routing can preserve strong service levels, while reinforcement learning can adapt routing choices to optimize broader reward trade-offs and manage disruption exposure.

Research Summary

This dashboard supports a research framework for adaptive supply chain routing under uncertainty. The system integrates:

- live public freight-related economic indicators,
- live weather disruption signals,
- stored research datasets for reproducible experiments,
- and routing policy evaluation using baseline, heuristic, and reinforcement learning approaches.

The goal is to study how AI-based policies balance cost, delivery reliability, and disruption resilience under dynamic operating conditions.